Introduction to the Cypher Training School and Workshop

Alberto Procacci

The program of the training school

- 6 hours from 9:00-12:00 and 14:00-17:00.
- Lunch breaks from 12:00-14:00.
- Coffee breaks of 15 minutes at 10:15 and 15:15.

Social event:

- Monday 7/09 from 18:00-20:00 at Bros restaurant.

CYPHER
cost EUROPEAN COOPERATION IN SCIENCE & TECHNOLOGY
Funded by the European Union

The content of the training school

The goal of the training school will be to explore fundamental techniques for the development of reduced-order models of reacting flows.

Trainers:

- (Monday 07/09) Prof. Miguel Mendez: dimensionality reduction techniques and modal analysis.
- (Tuesday 08/09) Prof. Alice Cicirello: probabilistic and physics-informed machine learning.
- (Wednesday 09/09) Prof. Anh Khoa Doan: system identification for reactive dynamical systems.

Scientific challenge

Modern CFD simulations of reacting flows generate very large datasets and remain computationally expensive.

Reduced-order models aim to capture the dominant flame dynamics with a fraction of the computational cost.

During this week, you will learn the tools needed to build such models and apply them to a realistic combustion dataset during the hackathon.

The program of the hackathon

- Thursday morning: intro to Codabench and the hackathon.
- Thursday afternoon and Friday: Hackathon.
- Saturday morning: Awards, poster session and closing.
- Lunch breaks from 12:00-13:30.
- Coffee breaks of 15 minutes at 10:15 and 15:15.

Social event:

- Thursday 10/09 from 19:00 at CoHop brewery.

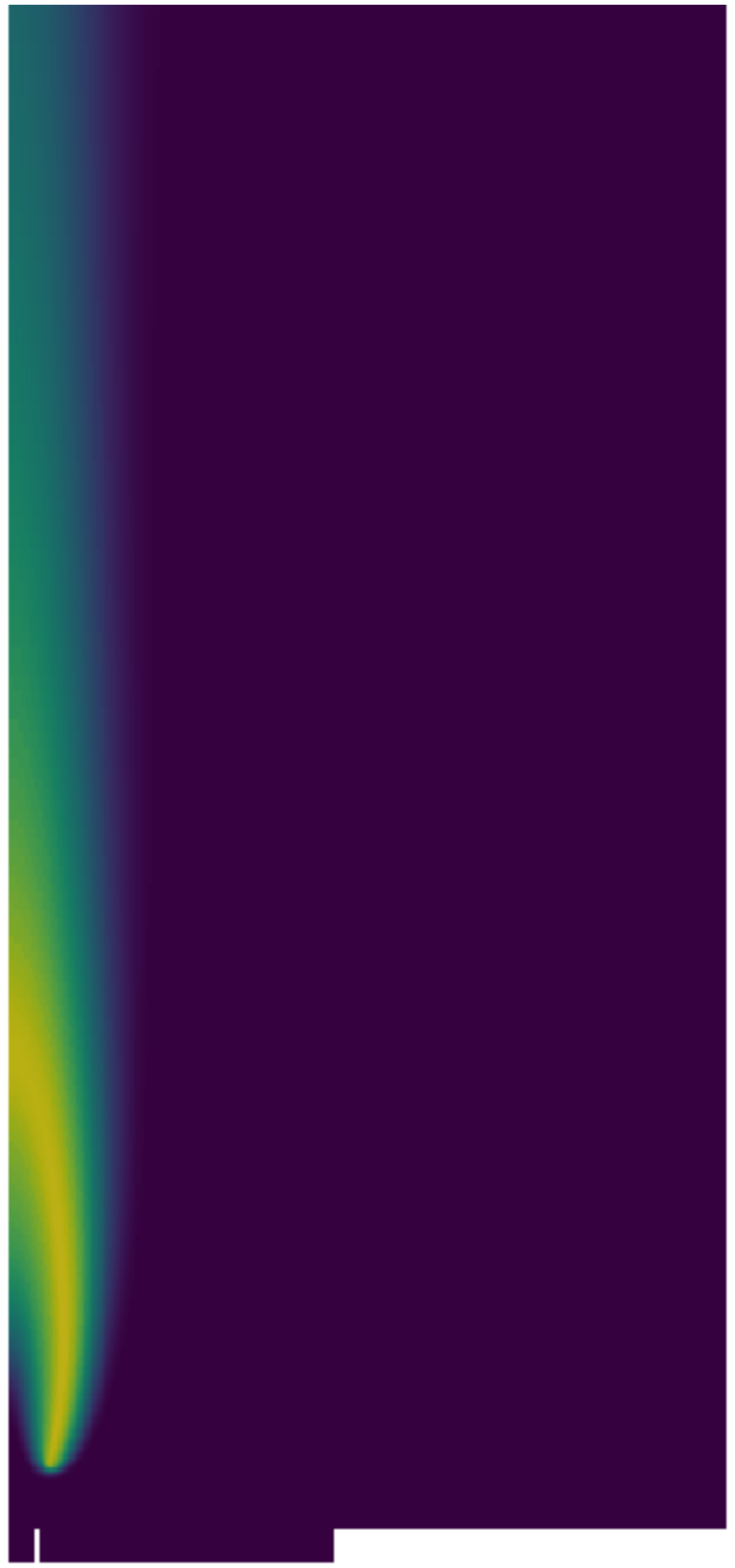
The content of the hackathon

Hackathon objective

Develop a reduced-order model capable of predicting the evolution of a reacting flow under unseen forcing conditions.

Training data will be provided for a flame excited over a broadband frequency range.

The submitted model will be evaluated on unseen step and sinusoidal excitations.



The groups

Group number	Member 1	Member 2	Member 3	Member 4
1	Tayyab Javaid	Emilio Mateo	Lu Tian	Tugba Gurler
2	Jason Chen	Jiajun Qiu	Pedro John	Mahwish Abbas
3	Ebrahim Rahmani	Kubilay BAYRAMOĞLU	Yihan YANG	Paul Kisza
4	Tamara Gammaidoni	Ahmet Aksoz	Arnob Kundu	Metin Millidere
5	Federico Risi	Sai Pavan Kumar Balabomma	Ghazanfar Mehdi	Vera Shiko
6	Luigi Gentile	Krešimir Kušić	Thomas Lawson	Maja Buljovcic
7	Michele Urbani	Qaisar Saddique	Mohamed Salah	Mustafa Yavuz
8	Prathmesh Choudhary	Constanza López	Luca Prattico'	Nicolò Del Santo

The groups have been created with the goal of balancing the python coding expertise.

CYPHER
cost EUROPEAN COOPERATION IN SCIENCE & TECHNOLOGY
Funded by the European Union

Resources

For the program, you can check the Cypher website:
https://cypher.ulb.be/activities/

For the hackathon, we have a dedicated Github repository:
https://github.com/burn-research/Cypher-challenge-dynamic-rom

and a Codabench challenge:
https://www.codabench.org/competitions/18039/

Resources

Link to Prof. Cicirello's Github repo:

https://github.com/mvulab/Cypher_day2_Cicirello_PML_and_PEML

CYPHER

EUROPEAN COOPERATION IN SCIENCE & TECHNOLOGY

Funded by the European Union

Hackathon

Hackathon objective

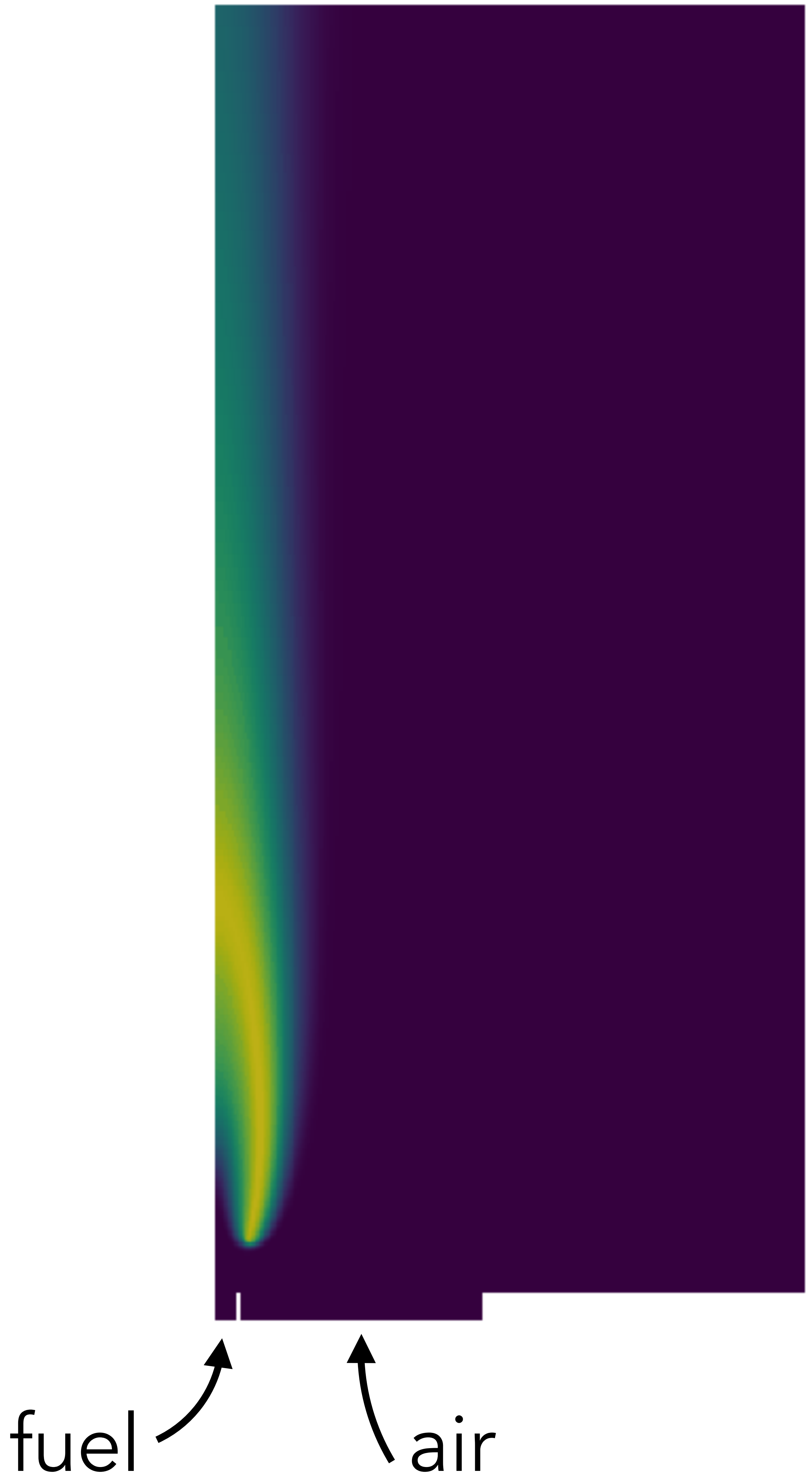
Develop a reduced-order model capable of predicting the evolution of a reacting flow under unseen forcing conditions.

Training data:

2D axisymmetric non-premixed laminar coflow flame.

The fuel is nitrogen-diluted methane (65% $\mathrm{CH_4}$ / 35% $\mathrm{N_2}$)

The features are $p, u_r, u_z, \rho, T, Q, Y_{\mathrm{CH_4}}, Y_{\mathrm{O_2}}, Y_{\mathrm{H_2O}}, Y_{\mathrm{CO_2}}, Y_{\mathrm{OH}}$



Hackathon

Boundary conditions (training):

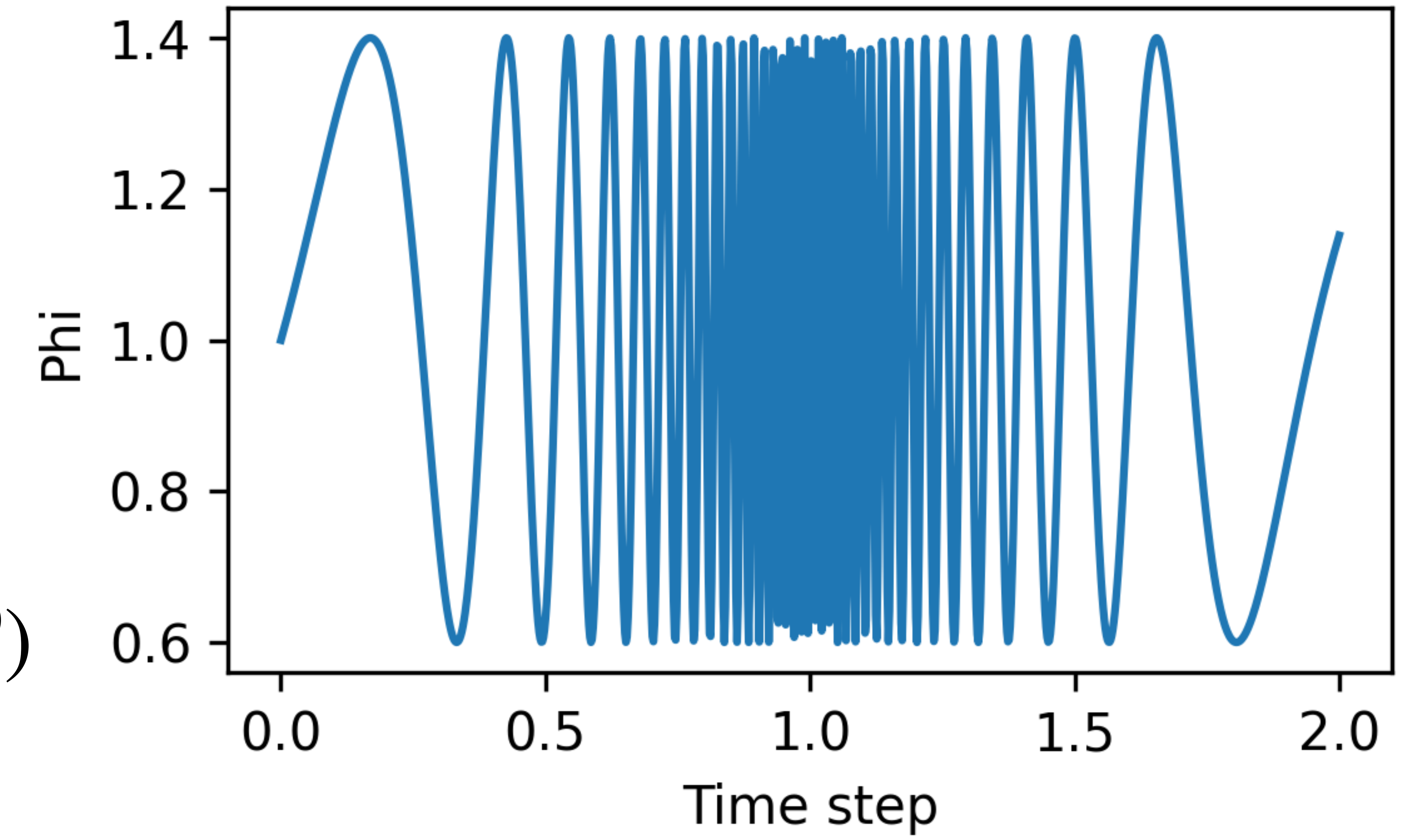
Time-varying boundary condition (sine sweep):

$u_f(r, t) = u_{\text{base}}\phi(t)(1 - r^2/R^2)$ where

$\phi(t) = 1 + A\sin(\psi(t))$ and

for $0 \leq t < T : \psi(t) = (2\pi f_0/k)(e^{kt} - 1)$

for $T \leq t \leq 2T : \psi(t) = \psi(T) + (2\pi f_1/k)(1 - e^{-k(t-T)})$



The sine sweep spans frequencies from 1 Hz to 80 Hz.

Hackathon

Boundary conditions (validation and test):

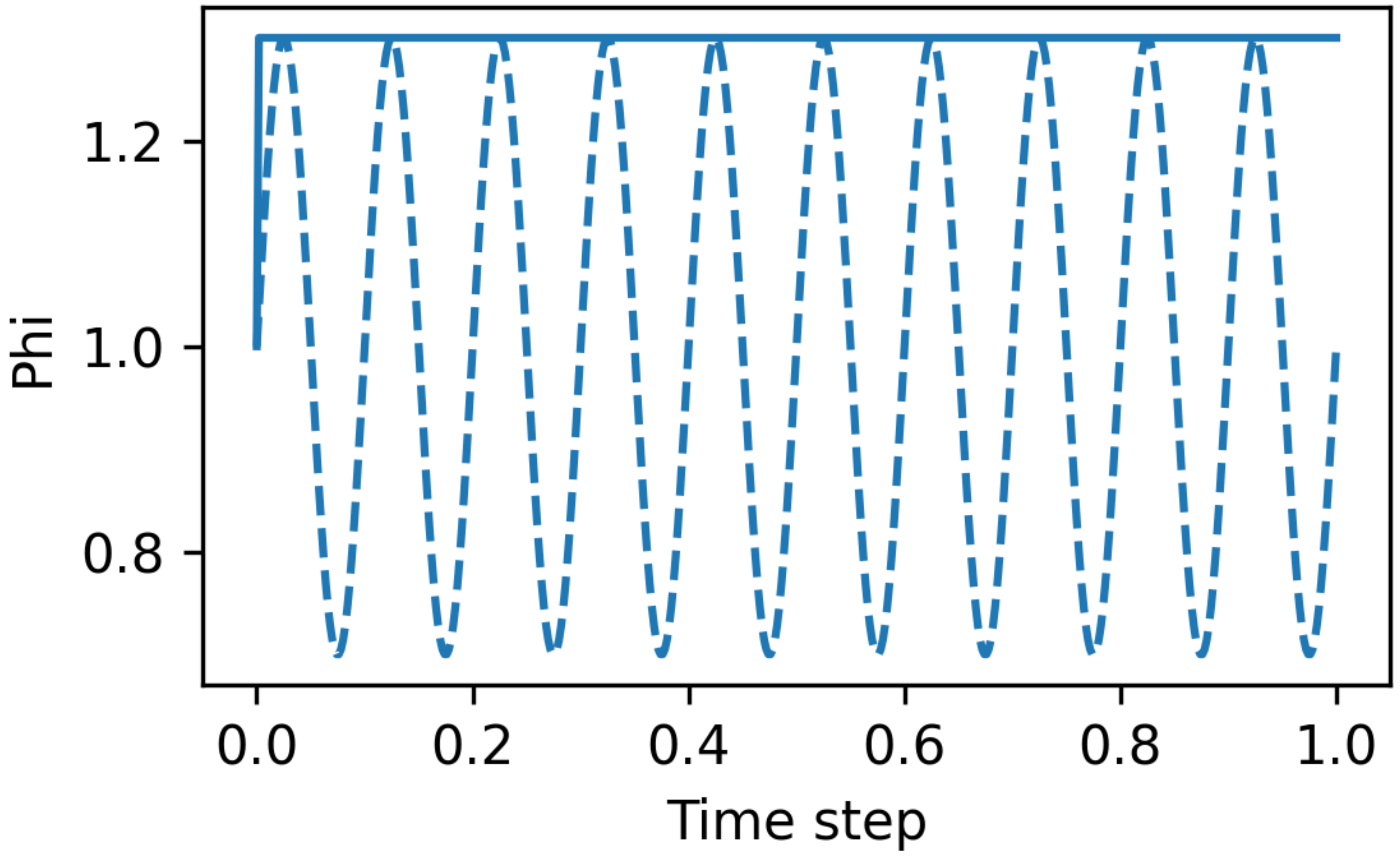
The validation and test cases are a collection of step and sinusoidal excitation with different amplitude.

Step:

$$\phi(t) = 1 + A$$

Sinusoidal:

$$\phi(t) = 1 + A(\sin(2\pi f t))$$



Hackathon

Scoring:

The scoring is divided into 3 parts: accuracy, physical consistency and inference time.

Accuracy: $e_{\mathrm{NRMSE}} = \dfrac{\sqrt{\frac{1}{NT}\sum_{i=1}^{N}\sum_{k=1}^{T}\left(Y_{i,k} - \hat{Y}_{i,k}\right)^2}}{\sigma_Y}$

Physical consistency: $e_{\mathrm{gain}} = \dfrac{|G - \hat{G}|}{G}$ with $G = \left|\mathscr{F}\{Q\}(f_0)\right|$, $\hat{G} = \left|\mathscr{F}\{\hat{Q}\}(f_0)\right|$,

$e_{\mathrm{phase}} = \dfrac{|\Delta\phi|}{\pi}$ with $\Delta\phi = \arg\left(e^{i(\hat{\phi}-\phi)}\right)$, $\phi = \arg\left(\mathscr{F}\{Q\}(f_0)\right)$, $\hat{\phi} = \arg\left(\mathscr{F}\{\hat{Q}\}(f_0)\right)$

Hackathon

Scoring:

The scoring is divided into 3 parts: accuracy, physical consistency and inference time.

Time: $e_{time} = 1/(1 + e^{-k(t-\tau)})$ with $k = 1$ and $\tau = 10$.

The error is transformed into a bounded score $S = \dfrac{1}{1 + e}$

The overall score is:

$$S = 0.5S_{\mathrm{NRMSE}} + 0.25S_{\mathrm{gain}} + 0.15S_{\mathrm{phase}} + 0.1S_{\mathrm{time}}$$

Codabench

Codabench is a platform which offers utilities to run data challenges.

To do:

- create an account,
- join the data challenge,
- follow the instructions to develop your model,
- submit the model.

The platform will automatically train the model and score it against the test data.

Example: operator inference

We have seen that we can use POD-Galerkin to project non-linear PDEs (Navier-Stokes) in the POD reduced space:

$\dot{\mathbf{a}} = \mathbf{c}_0 + \mathbf{L}\mathbf{a} + \mathbf{C} : (\mathbf{a} \otimes \mathbf{a})$ with $\mathbf{a}(0) = \mathbf{\Phi}^T \mathbf{W}_s \mathbf{v}(0)$

This can be rewritten as $\dot{\mathbf{a}} = \mathbf{O}d(\mathbf{a})$, where $\mathbf{O}$ is a linear operator and $d(\mathbf{a})$ is a quadratic function of $\mathbf{a}$.

What if we are too lazy to compute $\mathbf{O}$? We can try to estimate it from data:

$$\hat{\mathbf{O}} = \arg\min ||\dot{\mathbf{y}} - \mathbf{O}d(\mathbf{y})||_2^2$$

where $\mathbf{y}, \dot{\mathbf{y}}$ are observations of $\mathbf{a}, \dot{\mathbf{a}}$.

CYPHER
cost EUROPEAN COOPERATION IN SCIENCE & TECHNOLOGY
Funded by the European Union

Operator inference workflow

- Collect the snapshots in the data matrix (*A single data matrix? One per feature?*).
- Center and scale the data matrix (*Which scaling coefficient?*).
- Compute the POD (*What about the inner product?*).
- Compute the time derivative of $\mathbf{a}$ (*Finite differences? Which order?*).
- Compute the regression to estimate $\mathbf{O}$ (*Least-squares? Regularization? Probabilistic?*).
- Do not forget to include the effect of the external excitation (*Linear interaction?*).
- Integrate in time (*Explicit? Implicit?*).
- Project the solution back in the original space.

CYPHER
cost
EUROPEAN COOPERATION IN SCIENCE & TECHNOLOGY
Funded by the European Union

To keep in mind

- Memory in Codabench is limited, run your model with:

```
/usr/bin/time -l python model.py
```

  your peak RAM usage should be below 10 Gbs.
- We don't have access to GPUs.
- Training is limited to 6h, but better aim for something much shorter.
- Your submission could fail randomly, in that case try to re-upload it.
- As a backup, we will run the models locally.
- Indicate the number of your group in the submission.
- The pyhton libraries installed on Codabench are limited, use only the one indicated in the requirements.txt

The award

The winning group will receive a certificate for the best model and a mystery box